A descriptive analysis of diagnostic poultry antimicrobial resistance data from the Canadian AMRNet-Vet program

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Background

- Antimicrobial resistance (AMR) is a One Health concern with significant implications for animal, human, and environmental health
- In Canada, national AMR trends are monitored through programs such as AMRNet-Vet, which compiles diagnostic laboratory data from veterinary and animal health laboratories nationwide
- These data enable characterization of AMR patterns in key commodities, such as poultry, and support antimicrobial use (AMU) stewardship efforts
- Distinguishing poultry species and commodities (i.e., chicken vs. turkey, broilers vs. layers), improves interpretation of AMR patterns given differences in lifespan, disease pressures, and AMU

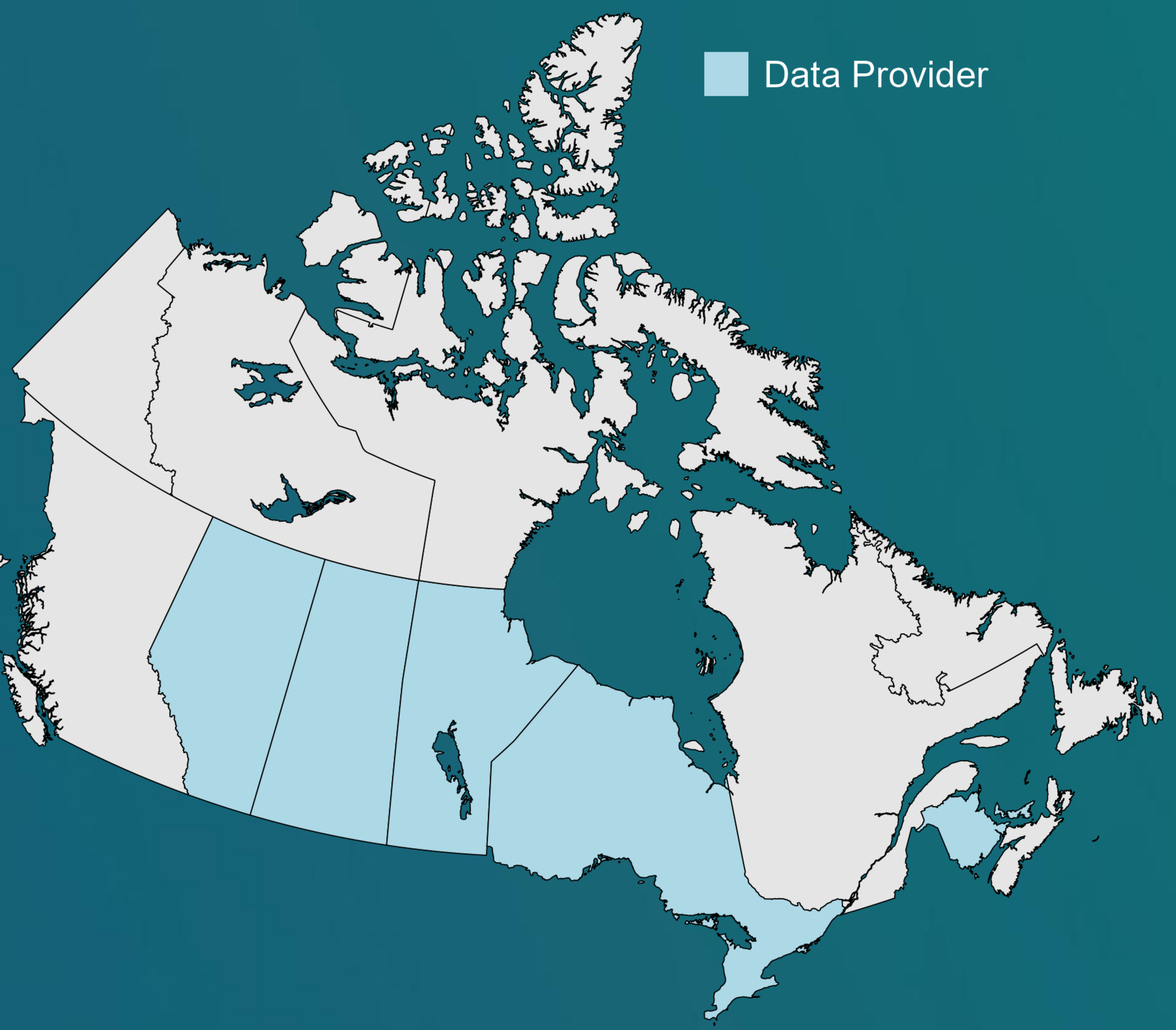
Objective

Summarize poultry bacterial recovery and antimicrobial susceptibility testing (AST) results reported to AMRNet-Vet from 2020–Mar 2025 and highlight the uses of these data in a One Health context while identifying opportunities to support continued data improvement.

Methods

- Data originated from 6 veterinary/animal health laboratories located in:
  - Alberta, Saskatchewan, Manitoba, Ontario, New Brunswick, and Prince Edward Island
- Between 2020-Mar 2025 (minimal 2025 samples were included in the 2020-2024 data pull), information from 20,440 poultry samples were submitted to AMRNet-Vet
- 64 unique bacteria were isolated often with multiple bacteria identified per sample
- Fourteen bacterial species were prioritized for further assessment due to their public health and clinical relevance in poultry:
  - *Avibacterium, Bordetella, Campylobacter, Clostridium, E. coli, Enterococcus, Gallibacterium, Klebsiella, Pasteurella, Proteus, Pseudomonas, Salmonella, Staphylococcus,* and *Streptococcus*
- Due to lack of AMR data and limited samples, *Clostridium* and *Campylobacter* were removed from further analysis
- *Bordetella* was removed from AMR analysis due to insufficient sample size (8 isolates)

Data Sources



Sample Breakdown

Results

- Although multiple species were reported (e.g., chicken, turkey, duck), most submissions were linked to unknown or unspecified poultry species (95%) making interpretation of the results difficult
- Among specimens with a known type, bone/bone marrow (18%), yolk sac (17%) and spleen (12%) were most common; however, specimen type was unreported for most samples (87%).
- Among the 11/14 identified bacteria:
  - AST practices varied greatly, with 65 unique AMs within 14 classes tested
  - Individual isolates were tested against 11 to 52 unique antimicrobials
- Following data sharing rules, AMR results were available for 11 of the identified bacteria and 7 antimicrobials (AMs) within 6 AM classes across the selected bacteria

Bacteria		Antimicrobials (% resistant)						
	N (min-max)*	Amoxicillin clavulanic acid	Ampicillin	Benzylpenicillin	Erythromycin	Gentamicin	Tetracycline	Trimethoprim sulfamethoxazole
Avibacterium	30 (30-30)							33%
E. coli	16376 (14402-16189)		45%			35%	54%	19%
Enterococcus	830 (590-660)		4%	13%			71%	
*Gallibacterium***	263 (249-262)		42%	67%	90%		80%	30%
Klebsiella	115 (59-115)	IR	IR			17%	33%	4%
*Pasteurella***	171 (149-171)		38%	26%	83%		56%	15%
Proteus	37 (33-37)		16%			9%	IR	14%
Pseudomonas	631 (630-630)					3%		
Salmonella	583 (513-581)		8%			32%	32%	3%
Staphylococcus	1843 (1623-1834)			9%	3%	2%	13%	1%
Streptococcus	123 (97-118)						68%	14%

*Due to varying denominators by antimicrobial, N = total isolates received by AMRNet-Vet for bacteria; (min-max) = the lowest and highest denominators for reportable antimicrobials in this table

**For *Pasteurella* and *Gallibacterium* there are no resistance breakpoints for any of the AMs except Erythromycin. In these other cases, and % reported is % non-susceptible

IR = Intrinsic Resistance

- Tetracycline and trimethoprim-sulfamethoxazole were the most common AMs on an AST panel for the bacteria of interest, while amoxicillin/clavulanic acid and erythromycin were the least common.
- It can be noted that tetracycline, erythromycin, and ampicillin show the highest % of resistant isolates while trimethoprim-sulfamethoxazole and gentamicin show the lowest.
- Lack of available animal host species data limits interpretation and temporal analysis as poultry populations change from year to year

Conclusion

These findings highlight both the value and complexity of diagnostic animal health data. The results demonstrate the potential of the data to support more comprehensive analyses, complement routine surveillance, inform AMU stewardship, and align with international AMR surveillance initiatives for diseased animals. Continued improvements in data granularity and harmonization may enhance the overall value and interpretability of these analyses.

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